

GERD

Lower oesophageal sphincter (LES)

- A muscular ring at the junction of the oesophagus and stomach that prevents acid reflux.

Diaphragmatic crura

- Muscle fibers of the diaphragm that surround the oesophagus, aiding in its closure.

Oblique entry

- The slanted angle at which the oesophagus enters the stomach, adding a natural barrier to reflux.

Reflux oesophagitis

- Inflammation of the oesophagus due to acid reflux.

Sliding hiatus hernia

- A condition where part of the stomach pushes into the chest through the diaphragm, disrupting oesophageal functions.

Cardiomyotomy

- A surgical procedure to cut the muscles of the oesophagus to relieve difficulty in swallowing.

. Vagotomy

- Cutting of the vagus nerve to reduce stomach acid production, which may weaken the LES.

Ascites

- Accumulation of fluid in the abdominal cavity, increasing intra-abdominal pressure.

Anticholinergic drugs

- Medications that block nerve signals involved in muscle contractions and secretions, slowing gastric emptying.

Systemic sclerosis

- A rare autoimmune disease causing thickening and scarring of tissues, affecting the oesophagus.

Aminophylline

- A drug used to treat asthma that relaxes smooth muscles, reducing LES tone.

. Beta-agonists

- Medications that stimulate certain nerve receptors, used for asthma and other conditions, but relax LES.

Barrett's oesophagus

- A condition where the lining of the oesophagus changes due to chronic acid exposure, increasing cancer risk.

Odynophagia

- Painful swallowing.

Dysphagia

- Difficulty in swallowing, which can be temporary or persistent.

Aspiration pneumonia

- Lung infection caused by inhaling food, stomach acid, or saliva.

Endoscopy

- A diagnostic procedure using a flexible tube with a camera to examine the digestive tract.

Oesophagitis

- Inflammation of the oesophagus.

Barrett's mucosa

- Abnormal changes in the oesophageal lining due to acid reflux.

Biopsy

- Removal of a small tissue sample for microscopic examination.

Barium meal

- A radiological test where a patient swallows a barium solution to highlight the digestive tract on X-rays.

Bernstein test

- A diagnostic test where acid is introduced into the oesophagus to reproduce symptoms of reflux.

Ischaemic heart disease

- A condition caused by reduced blood supply to the heart.

Oesophageal motility studies

- Tests measuring muscle contractions and movement in the oesophagus.

H₂ receptor antagonists

- Medications (e.g., Cimetidine, Ranitidine) that reduce stomach acid production by blocking histamine receptors.

Proton Pump Inhibitors (PPIs)

- Drugs (e.g., Omeprazole, Pantoprazole) that block stomach acid production by inhibiting enzyme activity.

Adjunctive therapy

- Additional treatments to complement the primary therapy.

Metoclopramide/Domperidone

- Medications that enhance LES tone and speed up stomach emptying.

Dilatation

- A procedure to widen a narrowed section of the oesophagus.

Fundoplication

- A surgical procedure to wrap the stomach around the oesophagus to strengthen the LES.

Peptic ulcer

Anastomosis

- A surgical connection between two parts of the digestive system.

Meckel's diverticulum

- A congenital pouch in the ileum, which can cause ulcers or bleeding.

Submucosa

- A layer of tissue beneath the mucosa, containing blood vessels and nerves.

Helicobacter pylori

- A bacteria that infects the stomach lining, causing ulcers and inflammation.

Urease

- An enzyme produced by H. pylori that breaks down urea into ammonia, damaging the stomach lining.

NSAIDs

- Non-Steroidal Anti-Inflammatory Drugs like ibuprofen and aspirin that can cause stomach damage.

Cytoprotection

- The stomach's natural ability to protect its lining from acid damage.

. Zollinger-Ellison syndrome

- A condition causing excessive stomach acid production due to tumors in the pancreas or duodenum.

Pyloric sphincter

- A muscular valve controlling the flow of stomach contents into the duodenum.

Epigastrium

- The upper central part of the abdomen, just below the ribcage.

Water-brash

- Sudden excess salivation due to acid reflux.

Haematemesis

- Vomiting of blood, often due to bleeding ulcers.

Gastric outlet obstruction

- Blockage of the stomach's exit, preventing food from passing into the duodenum.

Barium meal

- A diagnostic X-ray test using barium to highlight abnormalities in the digestive tract.

Serum gastrin

- A blood test measuring gastrin hormone levels, often elevated in Zollinger-Ellison syndrome.

Urea breath test

- A non-invasive test for detecting *H. pylori* by measuring exhaled carbon dioxide after consuming a urea solution.

Antacids

- Medications that neutralize stomach acid to provide symptomatic relief.

Alginates

- Substances that form a protective "raft" over stomach contents, reducing acid reflux.

Parietal cells

- Cells in the stomach lining that secrete gastric acid.

Proton Pump Inhibitors (PPIs)

- Medications that block the enzyme (H^+/K^+ ATPase) responsible for stomach acid production.

Zollinger-Ellison syndrome

- A rare condition causing excessive stomach acid due to tumors in the pancreas or duodenum.

Prostaglandin Analogues

- Drugs that mimic prostaglandins, increasing stomach mucosal protection.

Misoprostol

- A synthetic prostaglandin used to prevent ulcers caused by NSAIDs.

Colloidal Bismuth Compounds

- Medications that coat ulcers, protecting them and also acting against *H. pylori*.

Sucralfate

- A complex salt that coats ulcers, shielding them from acid and enzymes.

H. pylori

- A bacteria that infects the stomach lining, causing ulcers.

Triple Therapy

- A treatment regimen combining a PPI, amoxicillin, and clarithromycin to eradicate *H. pylori*.

Quadruple Therapy

- An enhanced regimen adding bismuth and tetracycline to triple therapy for resistant cases.

Gastrectomy

- Surgical removal of part of the stomach.

Billroth I anastomosis

- A surgical procedure reconnecting the stomach to the duodenum after partial gastrectomy.

Vagotomy

- Surgical cutting of the vagus nerve to reduce stomach acid production.

Pyloroplasty

- Surgical enlargement of the pyloric sphincter to improve gastric emptying.

Gastroenterostomy

- A surgical procedure creating a connection between the stomach and the small intestine.

Glossitis

Glossitis

- Inflammation of the tongue, causing swelling and color changes.

Papillae

- Small, nodule-like structures on the tongue's surface responsible for taste and texture.

. Pernicious anaemia

- A type of anaemia caused by vitamin B12 deficiency due to poor absorption.

Oral lichen planus

- A chronic condition causing white patches, swelling, and sores inside the mouth.

Erythema multiforme

- A skin and mucous membrane disorder causing red spots and inflammation.

Aphthous ulcers

- Painful sores inside the mouth, also known as canker sores.

Pemphigus vulgaris

- An autoimmune disease causing blisters on the skin and mucous membranes.

Syphilis

- A sexually transmitted bacterial infection that can cause tongue lesions.

Corticosteroids

- Anti-inflammatory medications, such as prednisone, used to reduce swelling.

Antimicrobials

- Drugs that kill or inhibit the growth of bacteria, fungi, or other microorganisms.

Malabsorption syndrome

Malabsorption

- A condition where the body cannot absorb nutrients from food properly.

Chronic pancreatitis

- Long-term inflammation of the pancreas, leading to enzyme deficiency.

Cystic fibrosis

- A genetic disorder affecting the lungs and digestive system, leading to thick mucus and malabsorption.

Gastroenterostomy

- A surgical connection between the stomach and the small intestine.

Crohn's disease

- A chronic inflammatory bowel disease affecting the digestive tract.

Terminal ileum

- The last part of the small intestine responsible for bile acid absorption.

Stagnant loop syndrome

- Bacterial overgrowth in a part of the intestine due to stasis, leading to malabsorption.

Coeliac disease

- An autoimmune disorder where gluten consumption damages the small intestine lining.

Tropical sprue

- A condition causing malabsorption in people living or traveling to tropical areas.

Whipple's disease

- A rare bacterial infection that impairs nutrient absorption.

Giardiasis

- An intestinal infection caused by the parasite *Giardia lamblia*.

Radiation enteritis

- Damage to the intestine due to radiation therapy.

Lactase deficiency

- Inability to digest lactose due to low levels of the enzyme lactase.

Telangiectasia

- Small, widened blood vessels visible on the skin or mucosal surfaces.

Abetalipoproteinaemia

- A rare genetic disorder affecting fat absorption and transport.

. Lymphatic obstruction

- Blockage of the lymphatic system, leading to impaired nutrient transport.

Pericarditis

- Inflammation of the pericardium, the sac surrounding the heart.

Diabetes mellitus

- A metabolic disorder characterized by high blood sugar levels.

Hyperthyroidism

- Overactive thyroid gland producing excessive hormones.

Hypoparathyroidism

- Underactive parathyroid glands causing low calcium levels.

Emaciation

- Extreme thinness due to severe weight loss.

Pitting pedal oedema

- Swelling in the feet that leaves a dent when pressed.

Steatorrhoea

- Fatty, pale, and foul-smelling stools due to fat malabsorption.

Follicular keratosis

- A skin condition causing rough, bumpy patches.

Xerophthalmia

- Severe dryness of the eye due to vitamin A deficiency.

Keratomalacia

- Softening of the cornea due to severe vitamin A deficiency.

Tetany

- Muscle cramps and spasms due to low calcium levels.

Osteomalacia

- Softening of bones due to vitamin D deficiency.

Haemorrhagic tendencies

- Increased risk of bleeding due to vitamin K deficiency.

Angular stomatitis

- Cracks or sores at the corners of the mouth.

Cheilosis

- Cracking and scaling of the lips.

Macrocytic anaemia

- Anaemia with abnormally large red blood cells, often due to folic acid or B12 deficiency.

. Areflexia

- Absence of reflexes.

Cardiac arrhythmias

- Irregular heartbeat.

. Rickets

- Bone deformities in children due to vitamin D deficiency.

. Koilonychia

- Spoon-shaped nails often caused by iron deficiency.

Sudan III stain

- A dye used to detect fat in stool samples.

Schilling test

- A diagnostic test to identify the cause of vitamin B12 (cobalamin) deficiency.

Radio-labelled cobalamin

- Vitamin B12 tagged with a radioactive marker to track its absorption and excretion.

. Intrinsic factor

- A protein produced by the stomach necessary for vitamin B12 absorption.

Pancreatic enzymes

- Digestive proteins secreted by the pancreas, aiding nutrient breakdown.

. D-xylose test

- A test for carbohydrate absorption using a simple sugar (D-xylose).

Celiac sprue

- An autoimmune condition causing gluten intolerance and intestinal damage.

Tropical sprue

- A condition causing malabsorption, prevalent in tropical regions.

. Whipple's disease

- A rare bacterial infection affecting nutrient absorption in the intestines.

Crohn's disease

- A chronic inflammatory condition affecting the gastrointestinal tract.

Barium meal contrast radiography

- An imaging technique using barium to visualize abnormalities in the digestive system.

Steatorrhoea

- Fatty stools resulting from fat malabsorption.

Serological studies

- Blood tests to detect antibodies associated with autoimmune diseases.

Cholyl-¹⁴C-glycine

- A radioactive compound used to test bile acid metabolism.

Lactose H₂ test

- A test measuring hydrogen in breath to detect lactose intolerance or bacterial overgrowth.

Duodenal biopsy

- Sampling of tissue from the first part of the small intestine (duodenum) for diagnostic purposes.

Jejunal biopsy

- Sampling of tissue from the jejunum (middle part of the small intestine).

**Cholestyramine**

- A medication that binds bile acids to prevent their effects in bile acid deficiency.

Intramuscular B12

- Vitamin B12 administered through injection for effective absorption.

Electrolyte deficiency

- A lack of essential minerals like sodium, potassium, or calcium.

Acute Diarrhoea

Aetiology:

- **Aetiology:** The study of the causes of diseases.

Infectious:

- **Viral:** Related to viruses (germs that cause infection).
- **Bacterial:** Related to bacteria (microscopic organisms that can cause infections).
- **Protozoal:** Related to protozoa (single-celled organisms that can cause diseases).

Non-Infectious:

- **Inflammatory bowel disease:** Disorders causing inflammation (swelling) in the intestines, such as **ulcerative colitis** and **Crohn's disease**.
- **Metabolic:** Related to the chemical processes in the body.
- **Sepsis:** A serious infection that spreads throughout the body, often leading to organ failure.
- **Coeliac disease:** An autoimmune disease where the body's immune system attacks the intestines when gluten is eaten.

Clinical Features:

- **Steatorrhoea:** Fatty stools, which are difficult to digest.
- **Nutritional deficiencies:** Lack of essential vitamins or minerals in the body.
- **Anaemia:** A condition where there aren't enough red blood cells to carry oxygen throughout the body.
- **Pellagra:** A disease caused by a deficiency of niacin (vitamin B3).
- **Peripheral neuropathy:** Damage to nerves outside the brain and spinal cord, causing symptoms like tingling or numbness.
- **Irritability:** A state of being easily upset or angered.

Investigations:

- **Schilling test:** A test to check vitamin B12 absorption.
- **D-xylose test:** A test to check the ability of the small intestine to absorb sugars.
- **Endoscopy:** A procedure to look inside the body, especially the digestive tract.
- **Biopsy:** Removal of a small sample of tissue for testing.
- **Exocrine:** Related to glands that secrete fluids outside the body (like sweat or digestive juices).
- **Serological studies:** Tests to detect infections by identifying antibodies or antigens in the blood.
- **Breath tests:** Tests that detect certain diseases or conditions through the gases you exhale.

Treatment:

- **Pancreatic insufficiency:** When the pancreas doesn't produce enough digestive enzymes.
- **Replacement therapy:** Replacing missing nutrients or substances in the body.

Anti-microbial / Anti-diarrhoeal Treatment:

- **Anti-microbial:** Medications that kill or slow the growth of microbes like bacteria, viruses, or fungi.
- **Anti-diarrhoeal:** Medications that treat or stop diarrhoea.

Anti-motility Agents:

- **Anti-motility agents:** Drugs that reduce the movement of the intestines, helping control diarrhoea.
- **Loperamide:** A medication that reduces the frequency of diarrhoea.
- **Diphenoxylate atropine:** A drug used to treat diarrhoea by reducing bowel movements.
- **Racecadrol:** A newer drug that helps reduce water and sodium loss in the stool.

Diagnosis:

- **Culture:** A test where bacteria or viruses are grown from a sample to identify the infection.
- **Parasite:** A living organism that lives in or on another organism, causing harm to the host.

Dysphagia

- **Dysphagia:** Difficulty or discomfort in swallowing food.

Causes:

a. Mechanical:

- **Luminal:** Referring to the space inside a tube or organ, like the oesophagus.
- **Bolus:** A large chunk of food.
- **Foreign body:** An object or substance that is not normally part of the body.
- **Intrinsic narrowing:** Narrowing of the inside of a structure due to internal factors.
- **Strictures:** Abnormal narrowing of a passage in the body.
- **Malignancy:** Cancer or a cancerous growth.

- **Extrinsic compression:** Pressure from an outside source that affects an organ or structure.
- **Cervical spondylosis:** Age-related wear and tear of the spine in the neck.
- **Retropharyngeal abscess:** A collection of pus behind the throat, usually due to infection.
- **Aortic aneurysm:** A bulging or swelling in the aorta (the large artery that carries blood from the heart).

b. Neuromuscular:

- **Sjogren's syndrome:** An autoimmune disease that attacks moisture-producing glands.
- **Oral anaesthesia:** Loss of sensation in the mouth.
- **Striated muscle:** Muscle tissue that has a striped appearance under a microscope, such as the muscles involved in swallowing.
- **Bulbar paralysis:** Weakness of the muscles involved in swallowing and speech due to nerve damage.
- **Pseudobulbar paralysis:** Similar to bulbar paralysis but often caused by brain damage.
- **Myasthenia gravis:** A chronic condition that causes weakness in voluntary muscles.
- **Motor neuron disease:** A disease that affects the nerves controlling muscles.
- **Polymyositis:** Inflammation of multiple muscles, causing weakness.
- **Myopathies:** Diseases affecting muscle tissue.
- **Rabies:** A viral disease that affects the nervous system and can cause swallowing problems.
- **Tetanus:** A bacterial infection that causes muscle stiffness and spasms.

c. Disorders of smooth muscle:

- **Scleroderma:** A condition where the skin and connective tissues become thick and hard.
- **Myotonic dystrophy:** A genetic disorder that causes muscle weakness and difficulty relaxing muscles.
- **Oesophageal spasm:** Sudden, painful contractions of the oesophagus.
- **Achalasia cardia:** A condition where the lower oesophagus has difficulty relaxing, making it hard to swallow.
- **Chagas disease:** A tropical parasitic disease that can affect the heart and digestive system.

Clinical Features:

- **Nasal regurgitation:** The backward flow of food or liquid into the nose during swallowing.

- **Bulbar palsy:** Weakness of muscles used for swallowing and speaking, often due to nerve damage.
- **Hoarseness of voice:** A rough or raspy voice often caused by irritation or disease affecting the voice box.
- **Laryngitis:** Inflammation of the voice box (larynx), causing hoarseness or loss of voice.

Investigations:

- **Barium swallow:** A radiological procedure in which a person swallows a barium solution to help visualize the oesophagus in X-rays.
- **Upper GI endoscopy:** A procedure using a flexible tube to look inside the upper digestive system (oesophagus, stomach, and part of the small intestine).
- **Oesophageal motility studies:** Tests that assess how well the muscles in the oesophagus are functioning.

Treatment:

- **Nasogastric tube feeding:** Feeding through a tube inserted into the nose and down the oesophagus to the stomach.
- **Gastrostomy/jejunostomy:** Surgical procedures where a tube is placed directly into the stomach or small intestine for feeding.

Acute Pancreatitis

- **Acute:** Sudden onset or rapid development.
- **Pancreatitis:** Inflammation of the pancreas.

Causes

- **Gallstones:** Hard deposits of cholesterol or calcium that form in the gallbladder and can block bile ducts.
- **Triglycerides:** A type of fat found in the blood, high levels can lead to pancreatitis.
- **Infections:** When harmful microorganisms invade the body, causing illness.
- **Trauma:** Injury caused by physical damage like accidents or surgery.
- **Medications:** Drugs or medicines used to treat illnesses, some of which can cause side effects like pancreatitis.

Pathophysiology

- **Enzymes:** Proteins that help speed up chemical reactions in the body.
- **Self-digestion:** When the body's digestive enzymes begin to break down its own tissue.

Clinical Features

- **Jaundice:** Yellowing of the skin and eyes due to a buildup of bilirubin in the blood.
- **Tachycardia:** Abnormally fast heart rate.

Diagnosis

- **Serum amylase:** A digestive enzyme found in the blood, high levels suggest pancreatitis.
- **Lipase:** Another digestive enzyme found in the blood, also elevated in pancreatitis.
- **Imaging:** Medical tests (like ultrasound or CT scan) that create pictures of the inside of the body.

Treatment

- **Hydration:** The process of providing water and fluids to maintain bodily functions.
- **Analgesics:** Medications used to relieve pain.
- **Fasting:** The practice of not eating or drinking for a period of time.
- **Antibiotics:** Medications used to treat bacterial infections.

Complications

- **Pseudocyst:** A fluid-filled sac that can form in the pancreas, not lined by normal cells.

- **Sepsis:** A serious infection that spreads throughout the body, potentially causing organ failure.
- **Chronic pancreatitis:** Long-term inflammation of the pancreas, leading to permanent damage.

Prognosis

- **Recovery:** The process of regaining health or strength after an illness or injury.
- **Complications:** Problems that arise during or after treatment of a disease.